Experiment 2

System Planning and Architecture Design of E-Commerce Application

D15A: 04 & 31

1. Introduction to System Planning

E-commerce platforms enable buying and selling of goods and services over the internet. A well-planned system ensures smooth user experience, secure transactions, proper inventory control, and reliable order processing. Before development begins, it is essential to perform structured system planning and architectural design to reduce risks and implementation errors.

This experiment focuses on designing a scalable, secure, and modular architecture for an E-Commerce Application that supports customers, administrators, and external payment services.

2. Goals of Architecture Design

- Design a structured multi-layer architecture
- Ensure separation of concerns between UI, logic, and database
- Enable secure and reliable transactions
- Support scalability for increasing users and products
- Maintain data integrity and system availability

3. Detailed Requirement Analysis

3.1 Functional Requirements

These describe what the system should do.

- User Registration with validation
- Secure Login & Authentication
- Product Search with filters (price, category, rating)
- Add to Cart / Remove from Cart
- Wishlist Management
- Order Placement and Invoice Generation
- Online Payment Integration
- Order History and Tracking
- Admin Panel for Product & Inventory Management
- Sales Report Generation

3.2 Non-Functional Requirements

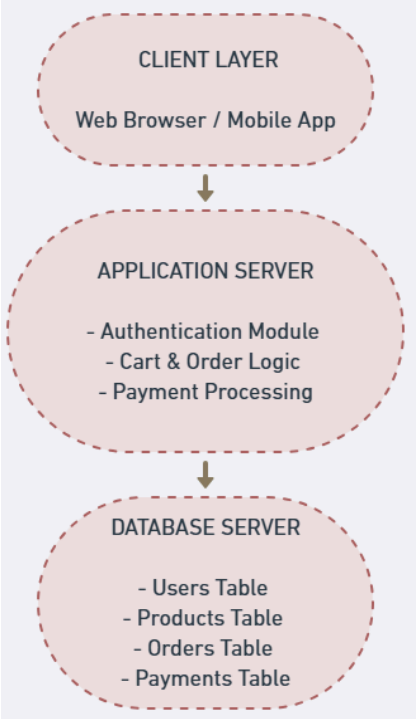
These describe how the system should perform.

- Fast response time (< 3 seconds page load)
- High availability (24/7 accessibility)
- Secure data transmission (HTTPS encryption)
- Reliable database backup
- User-friendly interface

4. High-Level Architecture Design (Color-Coded Representation)

Each layer performs a specific function and communicates only with adjacent layers, ensuring modularity and maintainability.

5. Three-Tier Architecture Diagram



This three-tier architecture improves security and performance by separating concerns.

6. Detailed System Modules

6.1 User Management Module

Handles user registration, login, password recovery, and profile management. It ensures authentication and authorization.

6.2 Product Management Module

Allows admin to add, update, delete, and categorize products. Maintains stock availability and pricing details.

6.3 Cart & Order Module

Manages selected products, calculates total price, applies discounts, and confirms orders.

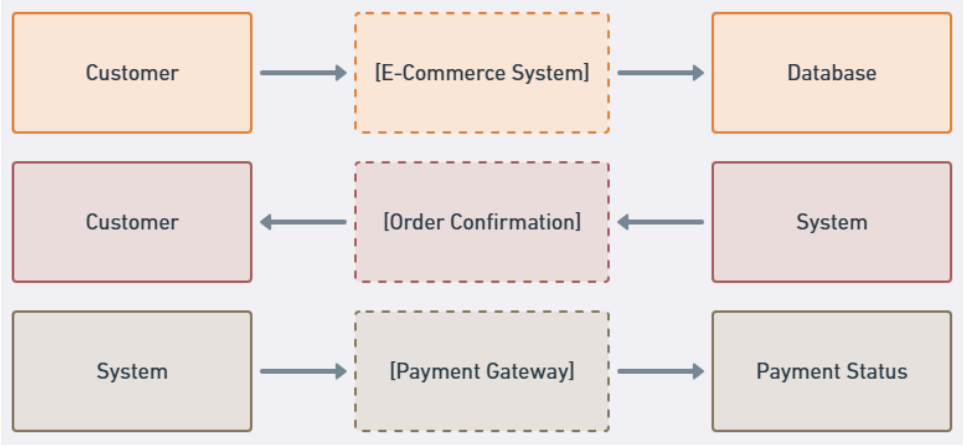
6.4 Payment Integration Module

Communicates with third-party payment gateways to process secure online transactions.

6.5 Notification Module

Sends order confirmation emails or SMS alerts after successful purchase.

7. Data Flow Diagram (Level 0)



This DFD shows how data moves between the customer, system, database, and payment gateway.

8. Sequence Diagram (Purchase Process)

User → UI : Select Product
UI → Server : Request Product Details
Server → Database : Fetch Data
Database → Server : Product Info
Server → UI : Display Product
User → UI : Checkout
UI → Payment Gateway : Payment Request
Payment Gateway → Server : Payment Confirmation
Server → Database : Store Order
Server → UI : Order Success Message

This diagram explains step-by-step interaction between components.

9. Database Design Planning

Key Tables

- Users (UserID, Name, Email, Password)
- Products (ProductID, Name, Price, Stock)
- Orders (OrderID, UserID, Date, TotalAmount)
- OrderDetails (OrderID, ProductID, Quantity)
- Payments (PaymentID, OrderID, Status)

ER Concept Diagram

User (1) —— *Places* —— (M) Order
Order (1) —— *Contains* —— (M) Product
Order (1) —— *Linked To* —— (1) Payment

10. Security Architecture Planning

- HTTPS protocol for encrypted communication
- Password hashing and secure authentication
- Role-based access control (Admin/User)
- SQL injection prevention
- Payment tokenization

Security is integrated at every layer of the architecture.

11. Scalability and Performance Strategy

- Modular service-based architecture
- Efficient indexing in database

- Load balancing (conceptual planning)
- Caching frequently accessed products
- Optimized server-side processing

12. Deployment Architecture (Conceptual)

User Device
▼
Web Server (Frontend Hosting)
▼
Application Server (Business Logic)
▼
Cloud Database Server
▼
External Payment API

This deployment structure ensures distributed and reliable operation.

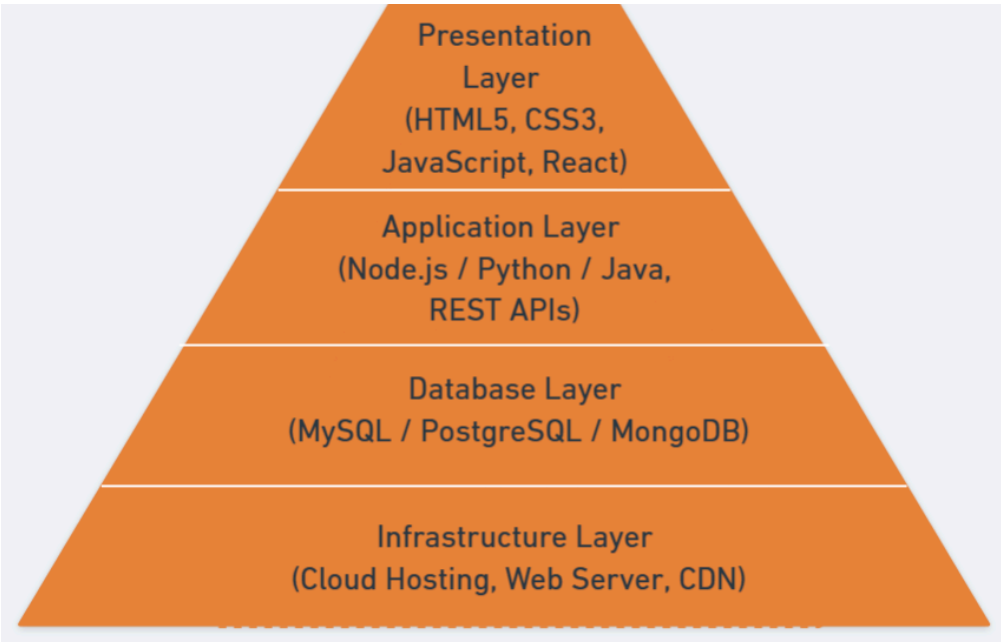
13. Development Planning Timeline

Phase	Activity	Description
Phase 1	Requirement Study	Identify system needs
Phase 2	Architecture Design	Create diagrams & models
Phase 3	Database Design	Define tables & relationships
Phase 4	Implementation	Coding modules
Phase 5	Testing	Unit & Integration Testing
Phase 6	Deployment	Hosting & Configuration

14. Tools and Technology Stack (Planned)

The selection of appropriate technologies is crucial for building a secure, scalable, and high-performance E-Commerce application. The technology stack is divided into layers based on functionality.

🔺 **Technology Stack Triangle (Layered Stack Representation)**



In this triangle structure, the upper layers depend on the lower layers. Each layer is responsible for a specific function in the overall architecture.

14.1 Presentation Layer

This layer is responsible for the user interface and user experience.

- HTML structures the web pages.
- CSS provides styling and responsive design.
- JavaScript or React enables dynamic interaction. It ensures smooth navigation, product browsing, and cart interaction.

14.2 Application Layer

This layer contains the business logic of the system.

- Handles user authentication and authorization.
- Processes cart operations and order placement.

14.3 Database Layer

This layer stores and manages application data.

- Stores user details, product information, and order records.
- Maintains data consistency and integrity.
- Supports fast retrieval using indexing techniques. It acts as the backbone of the e-commerce system.

14.4 Infrastructure Layer

This layer supports deployment and system availability.

- Cloud servers host the application.
- SSL ensures secure encrypted communication. It ensures reliability, scalability, and secure access.

This structured technology stack provides a balanced approach between performance, maintainability, and scalability.0

15. Conclusion

The System Planning and Architecture Design of the E-Commerce Application demonstrates how a structured, layered, and modular approach ensures scalability, security, and efficiency. By defining system components, workflows, database structure, and deployment strategy in advance, the application becomes easier to develop, maintain, and expand in the future.